Strand 4: Handling Data

Sub-strand 2: Chance and Probability

Lesson 1: Probability, Sample Spaces, and Dependent Events

1. Curriculum Standard and Overview

- **Strand**: 4 — Handling Data
- **Sub-strand**: 2 — Chance and Probability
- **Content Standard**: B9.4.2.1
- **Indicators**:
- **B9.4.2.1.1** — Perform a probability experiment involving two dependent events (without replacement).
- **B9.4.2.1.2** — Express probabilities as fractions, decimals, percentages, and ratios.
- **Mock-paper links**: P2 Q5(a) (pens in a bag: red, blue, green; probability as fraction/decimal/percentage/ratio; two pens without replacement).

2. Preamble — Chance in Everyday Language

"The chance of rain is 30%." "A coin has a 1 in 2 chance of heads."

Probability puts a **number** on how likely an event is:

`P(event) = (number of favourable outcomes)/(total number of possible outcomes)`

The result is always between **0 and 1** (or 0% and 100%):

- `0` = impossible
- `1` = certain

3. Basic Vocabulary

Word	Meaning
Experiment	an action with an uncertain result (toss a coin, pick a pen)
Outcome	one possible result
Sample space S	the set of all possible outcomes
Event	a collection of outcomes we care about

4. Probability in Different Forms

Mock P2 Q5(a): A bag has 5 red, 3 blue, 2 green pens. Total = `10`.

1. `P(red) = (5)/(10) = (1)/(2)`
2. `P(blue) = (3)/(10)`

- (α) decimal: `0.3`
- (β) percentage: `30%`
- (γ) ratio: `3 : 7` (favourable : not-favourable) or `3 : 10` (blue : total) — in BECE, write **3 : 7** when asked "as a ratio" of the event to its complement, or **3 : 10** if they mean blue to total. State which you mean.

Always simplify fractions.

5. Two-Stage Experiments

5.1 With replacement (independent)

The bag is unchanged between picks. Multiply the same probabilities.

5.2 Without replacement (dependent) — Mock P2 Q5(a)(iii)

If 2 pens are picked in succession **without replacement**, $P(\text{both green})$:

1. First pick: green pens $= 2$, total $= 10 \rightarrow P(G_1) = (2)/(10) = (1)/(5)$.
2. Second pick: green pens now $= 1$, total $= 9 \rightarrow P(G_2 \mid G_1) = (1)/(9)$.
3. Both green: $(1)/(5) \times (1)/(9) = (1)/(45)$.

Answer: $(1)/(45)$**.**

Key idea: *without replacement, the denominator* ***and*** *the numerator change after the first draw.*

6. Tree Diagrams (Preview)

```
G (2/10) → G (1/9) → P(GG) = 2/10 × 1/9 = 1/45
/
---<
\
not G (8/10) → ...
```

Along each branch multiply; across branches add.

7. Twenty Fully Worked Examples

Example 1 — Mock P2 Q5(a)(i)

$P(\text{red}) = (5)/(10) = (1)/(2)$.

Example 2 — Mock P2 Q5(a)(ii)

$P(\text{blue}) = (3)/(10) = 0.3 = 30\% = 3 : 7$ (or $3 : 10$).

Example 3 — Mock P2 Q5(a)(iii)

$P(\text{both green}) = (1)/(45)$.

Example 4 — Coin toss

$P(\text{heads}) = (1)/(2)$.

Example 5 — Fair die

$P(3) = (1)/(6)$; $P(\text{even}) = (3)/(6) = (1)/(2)$.

Example 6 — Certain event

$P(\text{a day has 24 hours}) = 1$.

Example 7 — Impossible event

$P(\text{rolling a 7 on a die}) = 0$.

Example 8 — Complement

$P(\text{not red}) = 1 - \frac{1}{2} = \frac{1}{2}$.

Example 9 — Cards

$P(\text{ace}) = \frac{4}{52} = \frac{1}{13}$.

Example 10 — Two coins (with replacement)

$P(\text{both heads}) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$.

Example 11 — Without replacement (letters)

Bag: A, A, B. Pick two without replacement. $P(\text{both A}) = \frac{2}{3} \times \frac{1}{2} = \frac{1}{3}$.

Example 12 — Percentage form

$P = \frac{3}{8} = 0.375 = 37.5\%$.

Example 13 — Ratio form

$P(\text{win}) = \frac{2}{7}$. Odds in favour $= 2 : 5$.

Example 14 — Spinner

Spinner with 4 equal sectors, 1 red: $P(\text{red}) = \frac{1}{4}$.

Example 15 — At least one

Die thrown twice. $P(\text{at least one six}) = 1 - P(\text{no six}) = 1 - \left(\frac{5}{6}\right)^2 = \frac{11}{36}$.

Example 16 — Mixed bag

Bag: 5 red, 3 blue, 2 green. $P(\text{not green}) = \frac{8}{10} = \frac{4}{5}$.

Example 17 — Two without replacement (red then blue)

$\frac{5}{10} \times \frac{3}{9} = \frac{1}{6}$.

Example 18 — Two without replacement (blue then red)

$\frac{3}{10} \times \frac{5}{9} = \frac{1}{6}$.

Example 19 — Order matters in calculation

$P(\text{red then blue}) = P(\text{blue then red})$ in this bag, but the working differs.

Example 20 — Word problem

20 pupils, 8 girls. $P(\text{girl}) = \frac{8}{20} = \frac{2}{5} = 0.4 = 40\%$.

8. Comprehensive Exercise Sets (40 Questions)

The authentic mock probability set (P2 Q5(a)) is included in full. Generated questions top up to 40. Spiral review ≈ 15%.

Exercise 1 — Basic probability (includes mock)

1. **[Mock 2026 P2 Q5(a)(i)]** In a pen bag, 5 are red, 3 are blue and 2 are green. What is the probability of picking a red pen?
2. What is `P(blue)` in the same bag?
3. What is `P(green)` in the same bag?
4. A fair coin is tossed. Find `P(heads)`.
5. **Spiral review:** The mean of 5, 8, 4, 7 and 2x is 6. Find x.

Exercise 2 — Different forms (includes mock)

1. **[Mock 2026 P2 Q5(a)(ii)]** Express the probability of picking a blue pen as a

(α) decimal; (β) percentage; (γ) ratio.

2. Express `(3)/(8)` as a decimal and a percentage.
3. If `P(win) = (2)/(7)`, write the odds in favour as a ratio.
4. **Spiral review:** `13x - 12 = 5x + 60`. Find x.
5. **Spiral review:** Area of a semicircle with `r = 7` (`π = 22/7`).

Exercise 3 — Without replacement (includes mock)

1. **[Mock 2026 P2 Q5(a)(iii)]** If 2 pens are picked in succession without replacement, what is the probability that both are green?
2. Bag: A, A, B. Pick two without replacement. Find `P(both A)`.
3. In the pen bag (5 red, 3 blue, 2 green), find `P(red then blue)` without replacement.
4. Find `P(blue then red)` without replacement.
5. **Spiral review:** If the area of a circle is 64π, find the circumference.

Exercise 4 — Sample spaces

1. List the sample space for two coin tosses.
2. List the sample space for one die roll.
3. `P(rolling a 7 on a die) = ?`
4. `P(a day has 24 hours) = ?`
5. **Spiral review:** Classify -3 under `\mathbb{N}, \mathbb{W}, \mathbb{Z}, \mathbb{Q}`.

Exercise 5 — Complements and "not" events

1. In the pen bag, find `P(not green)`.
2. `P(not red) = ?`
3. `P(not an ace) = ?` (from a standard deck)
4. **Spiral review:** Solve `2x + 1 > 9`.
5. **Spiral review:** `(-3)^(x-2) = -243`. Find x.

Exercise 6 — Two-stage independent

1. Two coins: find `P(both heads)`.
2. Two coins: find `P(two tails)`.
3. Two coins: find `P(one head, one tail)` in any order.
4. **Spiral review:** A van travels 154 km in `1(3)/(4)` h. Speed?
5. **Spiral review:** `(n)/(100) = 10.5`. Find n.

Exercise 7 — Real-life probability

1. A class has 20 pupils, 8 girls. Find `P(girl)` as a fraction, decimal, and percentage.

2. A bag has 6 blue and 4 red marbles. Find $P(\text{blue})$.
3. A spinner has 4 equal sectors, 1 red. Find $P(\text{red})$.
4. **Spiral review:** Find the mean of 3, 5, 7, 9.
5. **Spiral review:** Median of 20 marks from the stem-and-leaf (57.5) — verify the two middle values.

Exercise 8 — Challenge

1. Die thrown twice. Find $P(\text{at least one six})$.
2. In the pen bag, find $P(\text{two different colours})$ when two pens are drawn without replacement.
3. A card is drawn from a standard deck. Find $P(\text{ace or king})$.
4. **Spiral review:** In a market of 36 fishmongers, 28 sell mackerel, 18 tilapia, 13 both. How many sell neither?
5. **Spiral review:** Find the gradient through $(1, 2)$ and $(4, -4)$.

9. Full Step-by-Step Worked Solutions

Solutions to Exercise 1

1. $(5)/(10) = (1)/(2)$.
2. $(3)/(10)$.
3. $(2)/(10) = (1)/(5)$.
4. $(1)/(2)$.
5. $x = 3 \to$ **C**.

Solutions to Exercise 2

1. (α) 0.3; (β) 30%; (γ) $3 : 7$ (or $3 : 10$).
2. $0.375 = 37.5\%$.
3. $2 : 5$.
4. $x = 9$.
5. 77 cm².

Solutions to Exercise 3

1. $(2)/(10) \times (1)/(9) = (1)/(45)$.
2. $(2)/(3) \times (1)/(2) = (1)/(3)$.
3. $(5)/(10) \times (3)/(9) = (1)/(6)$.
4. $(3)/(10) \times (5)/(9) = (1)/(6)$.
5. 16π.

Solutions to Exercise 4

1. $\{HH, HT, TH, TT\}$.
2. $\{1, 2, 3, 4, 5, 6\}$.
3. 0.
4. 1.
5. ■, ■.

Solutions to Exercise 5

1. $(8)/(10) = (4)/(5)$.
2. $(1)/(2)$.
3. $(48)/(52) = (12)/(13)$.
4. $x > 4$.
5. $x = 7$.

Solutions to Exercise 6

1. `(1)/(4)`.
2. `(1)/(4)`.
3. `(2)/(4) = (1)/(2)`.
4. 88 km/h.
5. `n = 1050`.

Solutions to Exercise 7

1. `(2)/(5) = 0.4 = 40%`.
2. `(6)/(10) = (3)/(5)`.
3. `(1)/(4)`.
4. 6.
5. 10th value 57, 11th value 58; median = `57.5` ✓.

Solutions to Exercise 8

1. `1 - ≤ft((5)/(6))^2 = (11)/(36)`.
2. `1 - P(both same) = 1 - ≤ft((1)/(45) + (3)/(90) + (3)/(90)) = (10)/(15) = (2)/(3)` (see note).

Detail: P(GG)=1/45, P(RR)= (5/10)(4/9)=2/9, P(BB)=(3/10)(2/9)=1/15. Sum same-colour = 1/45+2/9+1/15 = 1/45+10/45+3/45 = 14/45. Different colour = 1 – 14/45 = 31/45.

3. `(4)/(52) + (4)/(52) = (8)/(52) = (2)/(13)`.
4. 3.
5. `m = -2`.

10. Authentic Mock Items in This Lesson

Ref	Question	Answer
P2 Q5(a)(i)	`P(red)`	`(1)/(2)`
P2 Q5(a)(ii)	`P(blue)` as decimal, %, ratio	`0.3 = 30% = 3:7`
P2 Q5(a)(iii)	Both green, without replacement	`(1)/(45)`

11. Lesson Summary

Skill	Rule
Basic probability	favourable ÷ total
Forms	fraction → decimal → % → ratio
Without replacement	reduce total **and** remaining favourable
Complement	`P(not A) = 1 - P(A)`
Two-stage (independent)	multiply
Two-stage (dependent)	multiply with updated numbers

Lesson prepared under `Grade9-Maths/Rules.md` *— plain English, British notation, full workings required.*